

AYUSH

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Education

Thapar Institute Of Engineering And Technology

Bachelor of Engineering in Computer Science and Engineering

Patiala, India

2023 – 2027

Experience

EXPERIMENTAL LEARNING CENTRE | Thapar University

June 2025 – July 2025

Research Intern

[ELC CENTRE](#)

- Collected and preprocessed large-scale Bengali speech data from talk shows, performing noise reduction, segmentation, and WAV-standard formatting for training.
- Built a speech-emotion recognition pipeline with MFCC-based feature extraction and evaluated multiple ML models, optimizing **CatBoost to 82% accuracy**.
- Developed modular Python workflows for preprocessing, feature engineering, model training, and evaluation using scalable and reproducible practices.

Projects

Solar Panel Efficiency Prediction | Python, XGBoost, LightGBM, CatBoost, Optuna, SHAP

- Built a high-performance regression pipeline using GPU-optimized gradient boosting models, improving prediction accuracy to ~90% for real-world solar efficiency data.
- Engineered advanced features (polynomial transforms, weather-irradiance interactions, sensor-based metadata), boosting model performance and stability across test folds.
- Implemented Optuna-based hyperparameter tuning and SHAP explainability to refine critical features, reducing model variance and enhancing interpretability for stakeholders.
- Designed a modular ML workflow with error handling, versioned experiments, and scalable data pipelines suited for production environments.

AI-Driven Resume Ranking System | NLP, Python, spaCy, Transformers, SQL, FastAPI

- Developed an automated resume-screening engine that extracts skills, projects, and experience using NLP preprocessing and transformer-based embeddings.
- Designed a scoring algorithm to rank resumes based on role fit, reducing manual screening time by 70%+ for bulk applications.
- Created an SQL-backed data storage pipeline to structure parsed resume data, enabling fast retrieval and analytics for HR workflows.
- Built REST APIs with FastAPI for seamless integration with hiring dashboards and ATS (Applicant Tracking Systems).

Achievements

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- Ranked Top 6% globally (63rd out of 1000+ teams) in the Zelestra X AWS ML Ascend Challenge (2nd Edition)
 - Secured 21st Global Rank in the 20-day Shell.ai Hackathon 2025 for building a high-performance hybrid ML pipeline to predict sustainable fuel blend properties.
 - Cleared SIH 2025 Internal Hackathon (Problem SIH25108), selected from 256 participating teams.

Technical Skills

Languages: Python, C/C++, SQL

Frameworks: Scikit-learn, XGBoost, LightGBM, CatBoost, PyTorch, Transformers, FastAPI,

Database: MySQL, MongoDB, PostgreSQL

Developer Tools: Git/GitHub, VS Code, Linux, API integration